

USB, POWER PATHS, CHARGING AND FILTERING

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USB Section:

- USB-C connector (J3) pins: VBUS, D+, D-, CC1, CC2, SHLD, GND.
- Diodes: D2, D6, D7 (D-TVS).
- Resistors: R12 (100K), R8 (3K3), R9 (5.1K), R11 (5.1K).
- Capacitors: C10 (10uF).
- LED: D4 (MSK4010).
- Jumper: A1 (JUMPER).

Power Regulation Section:

- 3.3V Regulator (U1): NCP167BMX330TBG.
- Input: VBUS.
- Output: 3.3V.
- Capacitors: C7 (1uF), C12 (1uF).

Battery Charging Section:

- Lipo Charger (U2): CHRG1 LP4055A.
- Current Limit Resistor (R10): 1.5A.
- Output Filter (C3): 10uF.
- Charging Pin: CHRG.
- Full Charge Pin: FULL.
- Ground Pin: GND.

Other Components:

- Resistor R1 (3K3).
- Diode D5 (MSK4010).
- Capacitor C4 (2.2uF).
- Resistor R3 (3K3).
- Capacitor C1 (10uF).

Notes:

- R3 Charge Rate:
 - 10K = 110mA
 - 5.0K = 200mA
 - 3.3K = 300mA**
 - 2.0K = 500mA
 - 1.2K = 830mA
- 830mA charge rate is the recommended fastest this PMIC can charge.
- SOLDERJUMPERCLOSED

BATTERY FUEL GAUGE

LD02 & AUTO-SHUTDOWN

The circuit diagram illustrates the connection of two integrated circuits, U2 and IC1B, to control the LD02 and AUTO-SHUTDOWN functions.

U2 (NCP167BMX330TBG) Connections:

- IN (Pin 4):** Connected to VBUS.
- EN (Pin 3):** Connected to LD02_EN.
- OUT (Pin 1):** Connected to 3V3_2.
- EXP (Pin 5):** Connected to GND.
- GND (Pin 2):** Connected to GND.

A 1µF capacitor (C17) is connected between VBUS and GND.

IC1B (NL175V08XV5T2GDRL) Connections:

- (Pin 1):** Connected to VDD_SPI.
- (Pin 2):** Connected to IO17.
- (Pin 4):** Connected to LD02_EN.
- (Pin 5):** Connected to GND.

A 1µF capacitor (C22) is connected between 3V3_2 and GND.

EXTERNAL FLASH & PSRAM

The image contains two circuit diagrams showing the connection of external memory components to a microcontroller via SPI.

Top Diagram: FLASH (W25Q128JVSQ)

- IC3:** W25Q128JVSQ
- Connections:**
 - CS (Pin 1):** Connected to SPICSS0 through a 100K resistor (R2) to VDD_SPI.
 - SQ(D1) (Pin 2):** Connected to SPIQ.
 - WP(D2) (Pin 3):** Connected to SPIWP.
 - GND (Pin 4):** Connected to GND.
 - VCC (Pin 8):** Connected to VDD_SPI.
 - HOLD(D3) (Pin 7):** Connected to VDD_SPI.
 - SCK (Pin 6):** Connected to SPICLK.
 - SI(D0) (Pin 5):** Connected to SPID.
 - SPIHD (Pin 7):** Connected to VDD_SPI.
- Capacitor:** C11 (0.1uF) connected between VDD_SPI and GND.

Bottom Diagram: PSRAM (PSRAM1N16080)

- IC7:** PSRAM1N16080
- Connections:**
 - CE (Pin 1):** Connected to SPICSS1 through a 100K resistor (R5) to VDD_SPI.
 - SI01 (Pin 2):** Connected to SPIQ.
 - SI02 (Pin 3):** Connected to SPIWP.
 - GND (Pin 4):** Connected to GND.
 - VCC (Pin 8):** Connected to VDD_SPI.
 - SI03 (Pin 7):** Connected to VDD_SPI.
 - SCLK (Pin 6):** Connected to SPICLK.
 - SI00 (Pin 5):** Connected to SPID.
 - SPIHD (Pin 7):** Connected to VDD_SPI.
- Capacitor:** C6 (0.1uF) connected between VDD_SPI and GND.

RGB LED & STEMMA/AT

The diagram illustrates the connection of an RGB LED to a WS2812B module and a STEMMA connector. The WS2812B module is shown with its pins: DI (Data In), DO (Data Out), GND (Ground), and VCC (3V3.2). The RGB LED is connected to the module's output pins: RED to DO, GREEN to GND, and BLUE to VCC. The STEMMA connector is shown with its pins: GND, VCC, SDA, and SCL. The connections are as follows: GND to GND, VCC to VCC, SDA to SDA, and SCL to SCL.

RGB LED

LED2

WS2812B

DI

DO

GND

VCC

3V3.2

STEMMA1

VERTICAL STEMMA CONNECTOR

STEMMA1	VERT
GND	1
VCC	2
SDA	3
SCL	4

IO18

IO9

IO8

3.3V

GND

HEADERS